

UFA, Russian Federation

WMO: 287220

Lat: 54.7109N

Lon: 55.8139E

Elev: 343

StdP: 14.51

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.7	-18.1	-29.1	1.1	-23.4	-23.1	1.6	-17.7	26.7	21.5	23.7	23.3	3.4	230	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	89.2	68.9	85.8	68.0	82.4	66.3	71.5	83.9	69.7	82.1	68.0	79.4	7.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	100.9	77.4	65.5	95.3	75.7	64.0	90.2	73.9	35.5	84.0	33.9	81.7	32.5	79.3	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.3	19.1	16.7	DB	-31.3	93.7	6.0	3.3	-35.6	96.1	-39.1	98.0	-42.5	99.9	-46.9	102.3	(4)
(5)			WB	-31.5	75.3	6.0	3.5	-35.8	77.8	-39.2	79.8	-42.6	81.8	-46.9	84.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.2	8.4	10.5	23.5	41.9	56.0	63.8	67.4	64.2	53.5	40.7	25.7	12.9	
	DBStd	23.50	13.83	13.00	9.68	9.81	8.64	7.56	6.23	7.31	7.69	8.19	11.32	13.99	
	HDD50	5773	1290	1105	821	281	40	4	0	1	48	305	729	1149	
	HDD65	9721	1755	1525	1286	693	298	111	48	107	352	753	1179	1614	
	CDD50	1837	0	0	0	38	227	419	540	442	154	17	0	0	
	CDD65	309	0	0	0	1	20	76	123	82	7	0	0	0	
	CDH74	3704	0	0	0	21	378	914	1359	920	112	0	0	0	
	CDH80	1225	0	0	0	2	92	306	461	340	24	0	0	0	
Wind	WSAvg	7.0	7.4	7.3	8.0	7.6	7.3	6.3	5.7	5.9	6.2	7.4	7.6	7.6	
Precipitation	PrecAvg	22.8	1.7	1.5	1.5	1.2	2.0	2.4	1.9	2.2	1.9	2.3	2.0	2.1	
	PrecMax	30.2	3.6	3.1	3.0	3.1	4.2	5.0	6.1	4.2	4.8	4.5	4.9	4.6	
	PrecMin	15.7	0.3	0.3	0.1	0.0	0.2	0.3	0.4	0.3	0.2	0.4	0.1	0.2	
	PrecStd	3.9	0.9	0.7	0.7	0.8	1.1	1.3	1.1	1.0	1.1	1.2	1.1	1.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.3	36.8	48.6	76.9	86.3	91.5	92.8	93.1	83.7	68.4	50.1	36.3	
		MCWB	33.7	34.0	40.8	57.2	63.9	68.7	70.2	68.9	62.5	53.4	47.2	34.0	
	2%	DB	32.5	34.0	41.9	70.4	82.6	87.6	88.1	88.1	76.9	62.7	44.8	34.5	
		MCWB	31.4	32.4	36.4	54.3	62.3	68.3	69.9	68.4	60.5	51.1	42.7	33.2	
	5%	DB	30.5	31.9	38.5	64.9	78.6	83.7	85.3	83.8	71.7	57.2	41.3	33.4	
		MCWB	29.1	30.4	34.5	51.3	60.5	66.9	69.0	67.5	58.8	49.1	39.3	32.4	
	10%	DB	28.0	28.6	35.9	58.9	73.5	79.2	82.0	78.8	67.3	52.7	38.4	30.8	
		MCWB	26.8	27.1	33.1	48.4	57.9	65.1	67.6	65.5	56.9	47.2	36.6	29.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.7	34.3	41.9	58.9	67.2	72.6	74.2	72.8	66.1	56.3	47.7	34.8
			MCDB	34.5	35.6	47.4	74.7	81.9	85.9	86.7	85.9	78.2	64.3	49.6	35.3
2%		WB	31.6	32.5	37.1	55.3	64.0	70.1	72.1	70.2	62.5	52.9	42.8	33.4	
		MCDB	32.4	33.6	40.6	68.0	78.5	82.8	84.4	83.6	72.9	60.0	44.1	34.2	
5%		WB	29.2	30.2	35.1	52.4	61.7	68.1	70.2	68.4	60.1	50.0	39.4	31.9	
		MCDB	30.3	31.6	37.5	63.5	74.9	80.1	81.9	80.7	68.9	55.2	40.9	32.8	
10%		WB	26.4	27.1	33.4	49.2	59.4	66.1	68.6	66.4	57.9	47.7	36.6	29.7	
		MCDB	27.7	28.6	35.4	58.7	71.3	77.3	79.6	77.3	65.3	52.4	37.9	30.8	
Mean Daily Temperature Range	5% DB	MDBR	15.0	18.0	18.3	19.9	23.1	21.9	21.9	21.5	19.5	13.9	11.0	13.3	
		MCDBR	11.4	13.3	17.6	29.1	31.2	28.6	28.0	29.0	27.5	23.9	10.9	9.8	
		MCWBR	11.1	11.9	12.8	16.2	14.5	13.0	12.6	13.1	14.0	14.4	9.5	9.3	
	5% WB	MCDBR	11.4	11.8	14.2	27.7	27.3	25.3	25.2	26.1	23.9	20.5	10.1	10.1	
		MCWBR	11.3	10.9	10.9	16.0	14.1	12.6	12.3	12.8	13.2	13.5	9.1	9.9	
Clear-Sky Solar Irradiance	taub	0.303	0.327	0.380	0.422	0.408	0.401	0.418	0.439	0.390	0.362	0.327	0.305		
	taud	2.024	2.002	1.927	2.057	2.245	2.322	2.279	2.196	2.297	2.263	2.159	1.996		
	Ebn at Noon	186	227	242	252	265	268	261	245	243	218	179	152		
	Edh at Noon	23	33	45	46	40	38	39	40	32	26	20	19		
All-Sky Solar Radiation	RadAvg	272	597	1054	1390	1695	1845	1823	1436	916	422	195	164		
	RadStd	31	57	95	155	158	201	174	147	99	94	44	37		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page